

Targeting the Wnt pathway in zebrafish as a screening method to identify novel therapeutic compounds

Joshua K Robertson, Kestral Danzmann, Sherise Charles, Katherine Blake, Annia Olivares, Solape Bamikole, Meghan Olson and Terence J Van Raay

Department of Molecular and Cellular Biology, University of Guelph, Guelph, ON Canada N1L 1A3

The first three authors contributed equally.

Corresponding author: Terence J Van Raay. Email: tvanraay@uoguelph.ca

Abstract

Activating mutations in the Wnt signaling pathway account for the initiation of greater than 90% of all colorectal cancers and this pathway has been implicated in numerous other diseases. Therefore, identifying small molecule inhibitors of this pathway is of critical importance towards identifying clinically relevant drugs. Numerous screens have been employed to identify therapeutic reagents, but none have made it to advanced clinical trials, suggesting that traditional screening methods are ineffective at identifying clinically relevant targets. Here, we describe a novel *in vivo* screen to identify small molecule inhibitors of the Wnt pathway. Specifically, treatment of zebrafish embryos with LiCl inhibits GSK3 kinase function, resulting in hyperactivation of the signaling pathway and an eyeless phenotype at 1 day post fertilization. Using the small molecule XAV939, a known inhibitor of Wnt signaling, we rescued the LiCl induced eyeless phenotype, confirming efficacy of the screen. We next tested our assay with 400 known small molecule kinase inhibitors, none of which should inhibit Wnt signaling below the level of GSK3 based on their known targets. Accordingly, none of these small molecules rescued the eyeless phenotype, which demonstrates the stringency of the assay. However, several of these small molecule kinase inhibitors did generate a non-Wnt phenotype in accordance with the kinase they targeted. Therefore, combining the efficacy, sensitivity, and stringency of this preliminary screen, this model will provide an alternative to the traditional *in vitro* screen, generating potentially clinical relevant drugs in a rapid and cost-effective way.

Keywords: Cancer, zebrafish, small molecules, Wnt signaling, LiCl, eyeless

Experimental Biology and Medicine 2014; 239: 169–176. DOI: 10.1177/1535370213514322

Introduction

Colorectal cancer (CRC) is the third most common cancer worldwide with over 1.2 million new diagnoses each year. What makes CRC unique among cancers is its dependence on a single signaling pathway. A recent comprehensive study reports that greater than 90% of all colon cancers are caused by activating mutations in genes within the Wnt signaling pathway.^{1–3} In parallel research, it is now well established that cancer stem cells (CSC) are the source for both the initiating tumor and recurrent malignancies that are therapy-resistant.^{4–7} Wnt signaling drives CSCs making this pathway the definitive signaling cascade that drives tumorigenesis in colon and many other cancers.^{8–16} Thus, the Wnt pathway in CSC is an excellent therapeutic target for a myriad of human cancers, particularly of the colon.^{5,9,10,17} To this end, there have been numerous small molecule screens for Wnt inhibitors, but these typically utilized cell lines with an artificial reporter. While such screens have had some success in identifying

Wnt inhibitors, none have advanced to stage III clinical trials.

To further our understanding of CRC development, we have developed an innovative strategy to use the *in vivo* stem cells of the zebrafish blastula in a semi-high-throughput screen for small molecules that target the Wnt pathway. We take advantage of the highly conserved Wnt signaling pathway,¹⁸ which is active in the stem cells of the early zebrafish blastula stage embryo.^{19–22} Zebrafish are rapidly becoming the preferred model for whole animal toxicity screens in the pharmaceutical industry^{23,24} as they are a well-established genetic tool for understanding the function of human genes in development and disease.^{25–27}

Wnt signaling pathway

In its resting state, in which there is no Wnt ligand (Figure 1(a)), an intracellular, constitutively active, destruction complex consisting of axin, adenomatous polyposis coli (APC) (both scaffolding proteins) and GSK3 (a kinase) binds